Monoleaflet Heart Valve Performance

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Introduction

Current mechanical heart valves pose a thrombosis risk, requiring lifelong anticoagulation. To address this, we developed a monoleaflet valve aimed at improving hemodynamics and reducing regurgitation. The design was evaluated through SolidWorks modeling, 3D prototyping, and in-lab testing using a mock circulatory loop (MCL) and Particle Image Velocimetry (PIV), though limitations exist in replicating in-vivo conditions.

Figure 1. MHV Design

Study objectives

- Designed and 3D printed a monoleaflet mechanical heart valve.
- Evaluated valve performance using a custom Mock Circulatory Loop (MCL).
- Used Particle Image Velocimetry (PIV) to assess flow dynamics and shear stress.
- **Hypothesis:** The monoleaflet mechanical heart valve will exhibit superior hydrodynamic performance compared to bileaflet valves, with lower regurgitation, reduced pressure drop, and more physiologic flow under simulated cardiac conditions.

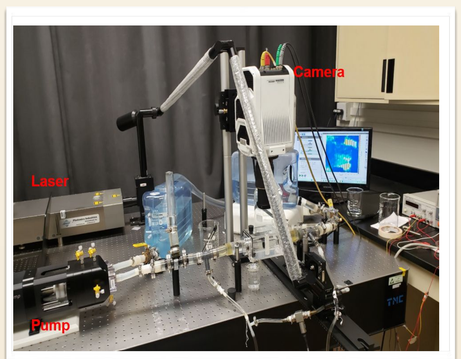

Figure 2. Set up of the MCL

Methods

In-vitro hemodynamic testing used a Mock Circulatory Loop (MCL) with Particle Image Velocimetry (PIV). A pulsatile pump and compliant tubing simulated physiological flow, while a laser and high-speed camera tracked particle movement to analyze velocity gradients during systole and diastole.

References

1. MacIsaac, S., Jaffer, I. H., Belley-Côté, E. P., McClure, G. R., Eikelboom, J. W., & Whitlock, R. P. (2019). How did we get here?: A historical review and critical analysis of anticoagulation therapy following mechanical valve replacement. Circulation, 140(23), 1933–1942. https://doi.org/10.1161/circulationaha.119.041105
2. Mayo Foundation for Medical Education and Research. (2023, September 27). *Aortic Valve Disease*. Mayo Clinic. https://www.mayoclinic.org/diseases-conditions/aortic-valve-disease/symptoms-causes/syc-20355117#dialogId1043509

Flow during Systole and Diastole for 50 mL/stroke and one at 70 mL/stroke

50 mL/stroke at 180 ppk

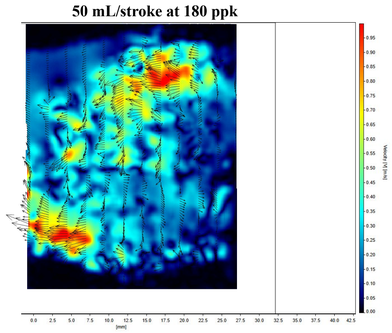

Figure 3. Early Systole

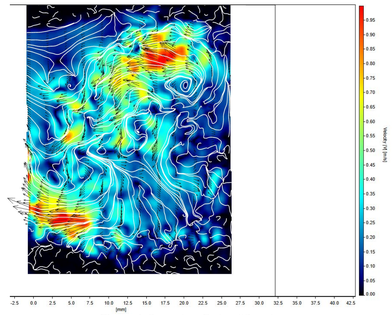

Figure 4. Streamline Early of Systole

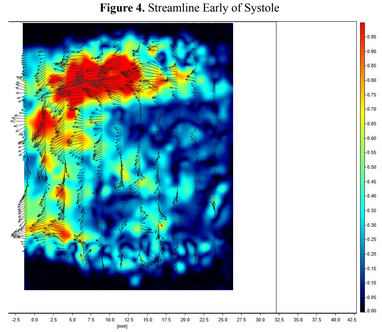

Figure 5. Late Systole

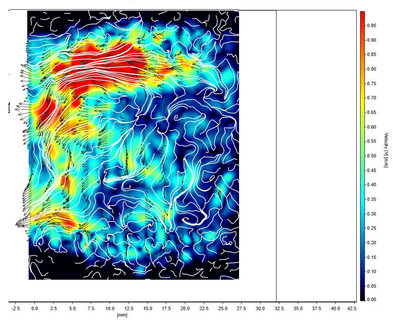

Figure 6: Steamline Late Systole

70 mL/stroke at 180 ppk

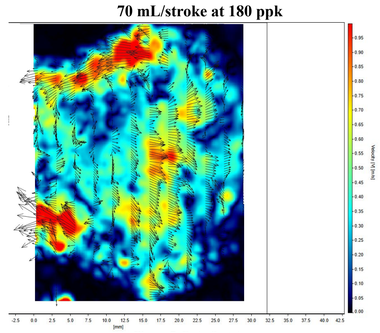

Figure 7. Early Systole

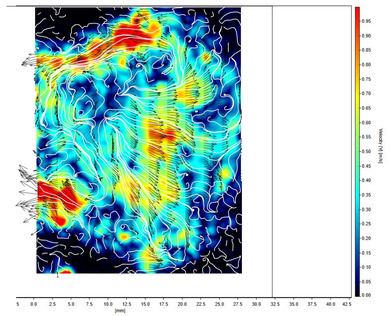

Figure 8. Streamline Early of Systole

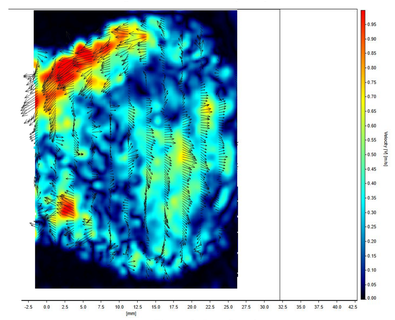

Figure 9: Late Systole

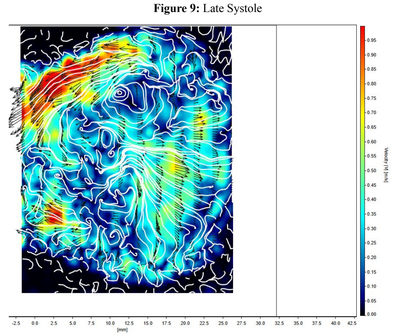

Figure 10: Streamline Late Systole

Major findings

Despite this limitation, the results indicate that the monoleaflet valve design has potential to improve hemodynamic performance. The valve was developed through multiple design iterations in SolidWorks and fabricated using a 3D printer with PVB filament. In-vitro testing and Particle Image Velocimetry (PIV) were used to validate flow characteristics and leaflet motion. The simplified structure supported smoother flow patterns, and the use of butterfly hinges with low-friction circular rods helped maintain stable and repeatable leaflet function.

$$\mathrm{EOA}(cm^2) = \frac{Q_{rms}(mL/s)}{51.6\sqrt{\frac{\Delta P(mmHg)}{\rho(g/cm^3)}}} \qquad RF = \frac{V_{retrograde}\ (mL/stroke)}{SV\ (mL/stroke)} \times 100\%$$

At 50 mL/stroke, 140 peak pressure:

- EOA = 0.337 ± 0.117 cm^2
- RF = 30.17 ± 0.19%

At 70mL/stroke, 180 peak pressure:

- EOA = 0.419 ± 0.119 cm^2
- RF = 19.70 ± 0.08%

Value target for EOA is ≥1.7 cm^2 and RF is ≤15% for a 27 mm diameter heart valve.
This shows that the valve has high resistance to flow as well as having high regurgitation.

Conclusions

The monoleaflet mechanical heart valve was tested using a mock circulatory loop (MCL) under controlled conditions with a stroke volume of 50 & 70 mL/stroke and a peak pressure of 180 mmHg (Figures 3–10). This setup was designed to evaluate the valve's hydrodynamic performance. Video recordings from laser-assisted flow visualization showed that the valve maintained consistent leaflet motion. However, residual regurgitation was observed, which likely influenced the pressure and flow data. These findings suggest the need for further improvements to the hinge design and leaflet sealing.

Future Work

- Utilize a high-resolution 3D printer to achieve finer details and smoother leaflet edges for improved flow performance.
- Substitute current printing materials with biocompatible or mechanically superior alternatives to better mimic physiological conditions.
- Optimize hinge design and reduce friction to enhance leaflet mobility and long-term durability.